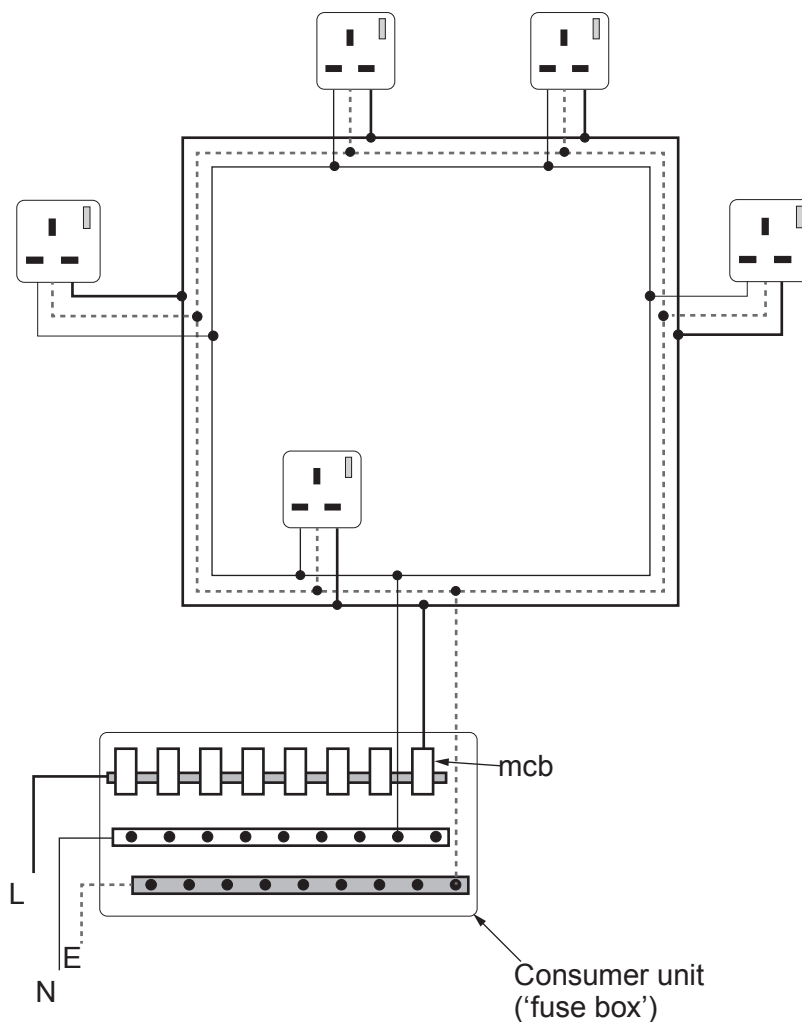


4. (a) The diagram shows a simple ring main circuit for the kitchen of a house. The sockets are used to supply a tumble dryer with a current of 9 A, a kettle with 10 A and a toaster with 5 A all at the same time.



- (i) Describe the function of the earth wire.

[2]

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- (ii) State **two** advantages of a miniature circuit breaker (mcb) over a fuse.

[1]

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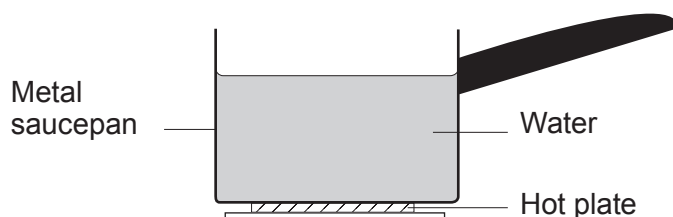
- (iii) The cable of the ring main can safely have 21 A passing through it. Explain why it is safe to use all of these appliances at the same time. [2]

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- (b) The diagram shows some water being heated in a metal saucepan.



Explain in terms of particles how heat is transferred to **all** of the water by conduction and convection. [4]

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